

Notes: Donaldson (AER, 2018)

Railroads of the Raj: Estimating the Impact of Transportation Infrastructure

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1 Big picture

- **Question.** How large are the economic gains from transportation infrastructure, and through what mechanism do those gains arise? The paper studies whether colonial India's railroad network lowered trade costs, expanded trade, and raised real income.
- **Setting.** The empirical setting is British India from 1870 to 1930, during which the railroad network expanded dramatically. The network rose from almost nothing in the mid-nineteenth century to more than 67,000 km of track by 1930.
- **Core data contribution.** Donaldson constructs a new archival district-level panel covering prices, agricultural output, daily rainfall, internal and external trade flows, and a digitized railroad network with segment-level opening dates.
- **Model-guided empirical design.** The paper is organized around four predictions from a many-region, many-commodity Ricardian trade model with costly trade: price gaps reveal trade costs in special cases, trade flows satisfy a gravity equation, railroads raise real income, and a model-based trade share is a sufficient statistic for welfare gains from trade-cost reductions.
- **Step 1: railroads and trade costs.** The paper uses salt varieties produced in unique locations to infer trade costs from spatial price gaps. Railroads sharply lower effective distance and therefore reduce trade costs.
- **Step 2: railroads and trade flows.** The paper estimates a gravity equation using internal trade flows across 47 trade blocks. Lower effective distance significantly increases trade flows.
- **Step 3: railroads and real income.** District railroad access is associated with a roughly 16 percent increase in real agricultural income per acre. Placebo lines that were surveyed or proposed but never built do not show comparable effects.
- **Step 4: mechanism.** A sufficient statistic from the trade model - the district's own-trade share - explains a little over one-half of the reduced-form income effect of railroads.
- **Economic takeaway.** The paper does not merely show that railroads correlate with growth. It links infrastructure to welfare through a quantified trade-cost and gains-from-trade mechanism, while also showing that other channels remain important.

2 Incremental contribution of the paper

- **From reduced-form infrastructure effects to mechanism.** Earlier infrastructure papers often estimate whether roads, dams, electricity, or railroads raise outcomes. Donaldson adds a structural trade model that explains why infrastructure should raise welfare through lower trade costs and newly exploited comparative advantage.
- **A new archival data contribution.** The paper digitizes and harmonizes colonial administrative records on output, prices, rainfall, trade, and railroads. The internal trade-flow data are especially valuable because within-country trade flows are rarely observed historically.

- **A network-based trade-cost measurement contribution.** The paper develops a method for inferring trade costs in a multimodal transport network. It combines special salt price gaps with a graph-theory lowest-cost-route algorithm and nonlinear least squares.
- **A bridge between trade theory and development history.** The empirical framework imports Eaton-Kortum-style Ricardian trade logic into a historical development setting, showing that comparative advantage and domestic market integration can explain a large part of infrastructure-driven income gains.
- **A sufficient-statistic contribution.** The paper empirically evaluates a sufficient statistic for welfare gains from trade-cost reductions: the share of expenditure a district sources from itself. This connects the reduced-form railroad effect to modern gains-from-trade formulas.
- **A credible-placement contribution.** The paper addresses nonrandom railroad placement using detailed historical placebo lines: proposed, reconnoitered, surveyed, Lawrence Plan, and Kennedy Plan lines that were never built.
- **A policy contribution.** Because the primitive shock is a reduction in trade costs, the analysis speaks not only to railroads in colonial India but also to roads, ports, logistics improvements, tariff reductions, and other trade-cost-reducing policies.

3 Data and measurement

3.1 District-level economic data

- The main panel covers Indian districts annually from 1870 to 1930.
- The welfare proxy is real agricultural income per acre, constructed from crop-level output, local crop prices, a local price index, and cultivated land area.
- The agricultural data cover 17 major crops: bajra, barley, cotton, gram, indigo, jowar, jute, linseed, maize, opium, ragi, rice, sesamum, sugarcane, tea, tobacco, and wheat.
- Agriculture is the natural focus because colonial India was predominantly agricultural and district-level nonagricultural output data are not comparably available.

Interpretation: The outcome is not modern GDP. It is the best available district-level welfare proxy for this historical setting. The focus on agriculture is both a data constraint and an economically meaningful choice because agriculture dominated the economy.

3.2 Prices and trade-cost data

- The paper uses retail prices for agricultural goods and, crucially, prices for six regionally differentiated types of salt.
- The salt varieties are useful because each type could only be produced in one source region, which allows spatial price gaps to reveal origin-destination trade costs under arbitrage.

- The salt data cover 133 districts in Northern India from 1861 to 1930.

Interpretation: Salt is not the main outcome of interest. It is a measurement device. Unique-source salt converts an otherwise unobserved trade-cost problem into an observed price-gap problem.

3.3 Trade-flow data

- The paper collects internal and external trade flows by rail, river, and sea.
- The gravity analysis uses 47 trade blocks and 15 commodities, including 14 agricultural commodities plus salt, from 1882 to 1920.
- Port trade blocks incorporate much international trade because goods moving through major ports can be traced in the trade records.

Interpretation: Observing internal trade flows is central because the paper wants to distinguish between railroads as mere local infrastructure and railroads as a force of domestic market integration.

3.4 Rainfall and productivity data

- The paper uses daily rainfall data from 3,614 meteorological stations.
- Rainfall is converted into crop-specific rainfall shocks using crop calendars that identify the sowing, growing, and harvesting periods for each crop and district.
- These crop-specific rainfall shocks are used as exogenous determinants of agricultural productivity in the trade model.

Interpretation: Rainfall variation is used not only as a control but also as a way to discipline the model's unobserved productivity terms.

3.5 Railroad network data

- The railroad network is digitized as a GIS map.
- Each approximately 20 km railroad segment is coded with the year in which it opened.
- District railroad access is coded as an indicator equal to one once any part of a district is penetrated by a railroad line.

Interpretation: The paper exploits both network geometry and timing. The timing gives district-level treatment variation; the geometry is needed to compute effective distances and trade-cost reductions.

4 Models and empirical specifications

4.1 Model environment

The paper builds a many-region, many-commodity Ricardian trade model. Regions are indexed by o or d , commodities by k , and varieties by j . Consumers have Cobb-Douglas preferences across commodities and CES preferences across varieties within each commodity:

$$U_o = \sum_{k=1}^K \left(\frac{\mu_k}{\varepsilon_k} \right) \ln \int_0^1 (C_o^k(j))^{\varepsilon_k} dj.$$

Productivity for variety j of commodity k in region o is drawn from a Frechet distribution:

$$F_o^k(z) \equiv \Pr(Z_o^k \leq z) = \exp(-A_o^k z^{-\theta_k}).$$

- A_o^k captures region-commodity productivity advantage.
- θ_k controls the dispersion of productivity draws and therefore the strength of comparative advantage.
- Trade costs are iceberg costs, denoted $T_{od}^k \geq 1$.

Interpretation: The model gives a tractable way to connect geography, productivity heterogeneity, trade costs, prices, trade flows, and welfare.

4.2 Result 1: price gaps and trade costs

For a homogeneous commodity that can only be produced in one origin region o , the log price gap between origin o and destination d equals the log trade cost:

$$\ln p_d^o - \ln p_o^o = \ln T_{od}^o.$$

The empirical trade-cost equation is

$$\ln p_{dt}^o = \beta_{ot}^o + \beta_{od}^o + \delta \ln \text{LCRED}(R_t, \alpha)_{odt} + \varepsilon_{odt}^o.$$

- $\text{LCRED}(R_t, \alpha)_{odt}$ is the lowest-cost-route effective distance between origin and destination, computed over roads, rivers, coasts, and railroads.
- R_t is the transport network open in year t .
- α contains mode-specific costs relative to rail.
- The model is estimated using salt price gaps and nonlinear least squares.

Interpretation: The empirical innovation is that geography enters through actual network routing rather than simple straight-line distance.

4.3 Result 2: gravity trade flows

The model implies a gravity equation for bilateral trade:

$$\ln X_{od}^k = \ln \lambda_k + \ln A_o^k - \theta_k \ln r_o - \theta_k \ln T_{od}^k + \theta_k \ln p_d^k + \ln X_d^k.$$

Using the estimated trade-cost function, the empirical version is

$$\ln X_{odt}^k = \beta_{ot}^k + \beta_{dt}^k + \beta_{od}^k - \theta_k \hat{\delta} \ln \text{LCRED}(R_t, \hat{\alpha})_{odt} + \varepsilon_{odt}^k.$$

Interpretation: Origin-year-commodity, destination-year-commodity, and origin-destination-commodity fixed effects absorb productivity, expenditure, prices, and time-invariant bilateral barriers. The coefficient on effective distance identifies how trade responds to trade costs.

4.4 Productivity estimation using rainfall

The paper relates unobserved productivity to crop-specific rainfall:

$$\ln A_{ot}^k = \kappa \text{RAIN}_{ot}^k.$$

It then estimates

$$\ln \tilde{X}_{odt}^k = \beta_{od}^k + \beta_{dt}^k + \beta_{ot}^k + \kappa \text{RAIN}_{ot}^k + \varepsilon_{odt}^k.$$

Interpretation: Rainfall is used to recover the productivity side of comparative advantage, which is needed to compute the model-based trade share in Step 4.

4.5 Result 3: railroads and real income

The reduced-form welfare equation is

$$\ln \left(\frac{r_{ot}}{\tilde{P}_{ot}} \right) = \beta_o + \beta_t + \gamma \text{RAIL}_{ot} + \varepsilon_{ot}.$$

- $r_{ot}/\tilde{P}_{ot}$ is real agricultural income per acre.
- RAIL_{ot} equals one once district o has railroad access.
- District fixed effects absorb permanent district characteristics.
- Year fixed effects absorb aggregate shocks.

Interpretation: This is the main reduced-form estimate of the effect of railroad access on local welfare.

4.6 Result 4: sufficient statistic for welfare gains

The model implies that real income can be written as

$$\ln W_o = \Omega + \sum_k \frac{\mu_k}{\theta_k} \ln A_o^k - \sum_k \frac{\mu_k}{\theta_k} \ln \pi_{oo}^k.$$

The term π_{oo}^k is the share of region o 's expenditure on commodity k that it sources from itself. The empirical sufficient-statistic equation is

$$\begin{aligned} \ln \left(\frac{r_{ot}}{\bar{P}_{ot}} \right) - \left[\sum_k \frac{\hat{\mu}_k}{\hat{\theta}_k} \hat{\kappa} \text{RAIN}_{ot}^k \right] &= \beta_o + \beta_t + \gamma \text{RAIL}_{ot} \\ &+ \psi \left[\sum_k \frac{\hat{\mu}_k}{\hat{\theta}_k} \ln \pi_{oot}^k(\hat{\Theta}, \text{RAIN}_t, R_t, L) \right] + \varepsilon_{ot}. \end{aligned}$$

Interpretation: If the trade model is the full mechanism, then the railroad coefficient should disappear once the model-based trade share is included. Empirically, it falls by more than one-half but does not disappear.

5 Identification and empirical design

5.1 What the design can identify

- The paper identifies whether railroad arrival is associated with changes in real agricultural income within districts over time, after controlling for district and year fixed effects.
- The model-based analysis identifies how much of this reduced-form effect can be accounted for by lower trade costs and increased gains from trade.
- The sufficient-statistic exercise is not a randomized experiment; it is a disciplined decomposition using estimated model primitives.

Interpretation: The strongest causal claim is the reduced-form effect of railroad access on real agricultural income, supported by historical placement arguments and placebo lines. The mechanism claim is model-based and quantitative.

5.2 Railroad placement and selection concerns

A natural concern is that railroads may have been built in districts that were already on different growth paths. Donaldson addresses this concern in three ways.

- **Historical institutional argument.** Military objectives were central to railroad planning, and the network nature of railroads made precise targeting of local growth difficult.

- **District and year fixed effects.** The baseline specification compares a district to itself before and after railroad access, net of aggregate year shocks.
- **Placebo railroad lines.** The paper studies more than 40,000 km of lines that were proposed, reconnoitered, surveyed, or included in alternative plans but never built. These unbuilt lines generally do not predict real income gains.

Interpretation: The placebo strategy is powerful because it compares actual railroad construction to lines that planners considered seriously enough to document or survey. If growth targeting drove the results, these near-built lines should also show effects.

5.3 Trade-cost identification

- The trade-cost estimation uses salt varieties that could only be produced in specific locations.
- Since salt was widely consumed, destination prices are observed across space.
- Origin-year fixed effects absorb source price shocks.
- Origin-destination fixed effects absorb time-invariant bilateral barriers.
- Time variation comes from the changing railroad network and its effect on lowest-cost routes.

Interpretation: This is a clever way to measure a normally unobserved object: the cost of moving goods across space.

5.4 Gravity identification

- The gravity equation uses bilateral trade flows across trade blocks and commodities.
- Origin-year-commodity and destination-year-commodity fixed effects absorb supply and demand shifters.
- Origin-destination-commodity fixed effects absorb persistent bilateral trade barriers.
- The coefficient is identified from changes in effective distance induced by railroad construction.

Interpretation: The gravity step tests whether the trade-cost reductions measured from salt actually translate into expanded trade in other commodities.

5.5 Limitations of the identification design

- Railroad access may capture related infrastructure or information technologies, such as postal and telegraph improvements that followed railroads.
- Real agricultural income per acre is not a complete welfare measure and does not directly measure nonagricultural gains or income volatility.

- The model abstracts from many channels, including migration, technology diffusion, fiscal effects, and financial development.
- The sufficient-statistic exercise depends on the model, the estimated trade-cost parameters, and assumptions about productivity and market structure.

Interpretation: The paper’s strength is not that it rules out every channel. Its strength is that it quantifies one central channel and shows that the channel explains a large share of the reduced-form effect.

6 Tables and figures

6.1 Table 1: Descriptive statistics

Read:

- Table 1 reports the main variables used throughout the paper: real agricultural income per acre, salt prices, crop-specific rainfall shocks, and agricultural exports by trade block.
- Real agricultural income per acre is available from 1870 to 1930 and is the core outcome used in the welfare regressions.
- Salt prices are available from 1861 to 1930 and are used to infer trade costs.
- Crop-specific rainfall shocks are available at a much higher observation count because rainfall is measured by crop, district, and year.
- Trade data are available from 1882 to 1920 and are used in the gravity analysis.

Economic message: The table previews the paper’s empirical architecture: salt prices identify trade costs, trade flows identify the gravity response, rainfall identifies productivity variation, and real agricultural income measures welfare.

6.2 Figure 1: The evolution of India’s railroad network, 1860–1930

Read:

- Figure 1 shows the decadal expansion of the railroad network from 1860 to 1930.
- The network begins with a few trunk connections and gradually fills in the interior of India.
- By 1930, the railroad network connects a large share of the subcontinent and reshapes the transport geography of colonial India.
- The figure is based on a GIS database in which each approximately 20 km segment is assigned an opening year.

	Number of observations	Beginning of available data	End of available data
Real agricultural income per acre (base year rupees)	7,086	29.96 (16.35)	40.41 (50.56)
Price of salt, all sources (current rupees per maund)	7,336	5.17 (1.49)	3.21 (0.54)
Crop-specific rainfall shock (meters)	120,462	0.75 (0.67)	1.29 (1.38)
Total agricultural exports per trade block (millions of 1870 rupees)	1,193	19.07 (45.68)	44.63 (58.76)

Notes: Values are sample means over all observations for the year and variable in question, with standard deviations in parentheses. Earliest beginning and latest end of available data are: 1870 and 1930 for agricultural output and real agricultural income; 1861 and 1930 for salt prices; 1870 and 1930 for rainfall; and 1882 and 1920 for trade data. Land area used to calculate income per acre is total cultivated area in first year of sample. A “maund” is equal to 37.3 kg and was the standardized unit of weight in colonial India. Total agricultural exports (aggregating across commodities and destination blocks) per trade block converted from quantities to values (in 1870 rupees) using all-India average prices in 1870. Data sources and construction described further in online Appendix A.

Table 1: Descriptive statistics.

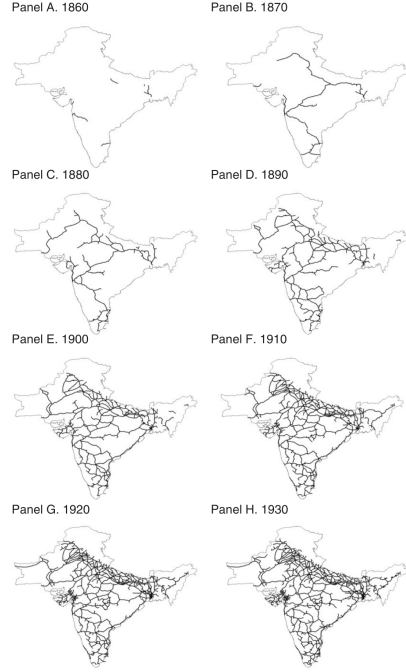


FIGURE 1. THE EVOLUTION OF INDIA'S RAILROAD NETWORK, 1860–1930

Notes: These figures display the decadal evolution of the railroad network (railroads depicted with thick lines) in colonial India (the outline of which is depicted with thin lines). The first railroad lines were laid in 1853. The figure is based on a GIS database in which each (approximately) 20 km long railroad segment is coded with a year of opening variable.

Figure 1: The evolution of India's railroad network, 1860–1930.

Economic message: The figure makes the source of identification visible. Districts become connected at different times as the network expands, generating variation in access, effective distance, trade costs, and real income.

6.3 Tables 2 and 3: Trade costs and trade flows

Read:

- Table 2 estimates the effect of effective distance on salt price gaps. The preferred nonlinear specification gives an elasticity of 0.169 for effective distance at estimated mode costs.
- The estimated mode costs are greater than rail for roads, rivers, and coast routes, implying that railroads reduce effective distance relative to alternative transport modes.
- Table 3 estimates the gravity response of trade flows. The pooled coefficient on log effective distance is around -1.603 , indicating that lower effective distance substantially increases trade.
- The interaction terms with commodity weight and freight class are not statistically important, supporting the use of salt-based trade-cost parameters for broader agricultural commodities.

Economic message: These two tables jointly establish the first half of the mechanism. Railroads reduce the cost of moving goods, and the reduction in trade costs generates a

substantial expansion in trade.

TABLE 2—RAILROADS AND TRADE COSTS: STEP 1

Dependent variable: log salt price at destination	(1)	(2)
log effective distance to source, along lowest-cost route (at historical freight rates)	0.088 (0.028)	
log effective distance to source, along lowest-cost route (at estimated mode costs)		0.169 [0.062, 0.296]
Estimated mode costs per unit distance: Railroad (normalized to 1)		1 N/A
Road		2.375 [1.750, 10.000]
River		2.250 [1.500, 6.250]
Coast		6.188 [5.875, 10.000]
Observations	7,345	7,345
R^2	0.946	0.946

Notes: Regressions estimating equation (12) using data on 6 types of salt (listed in online Appendix A), from 133 districts in Northern India, annually from 1861 to 1930. Column 1 and column 2 estimated by OLS and NLS respectively; both include salt type $\times$ year and salt type $\times$ destination fixed effects. "Effective distance to source, along lowest-cost route" measures the railroad-equivalent kilometers (because railroad freight rate is normalized to 1) between the salt source and the destination district, along the lowest-cost route given relative mode costs per unit distance. "Historical freight rates" used are 4.5, 3.0, and 2.25 respectively for road, river, and coastal mode costs per unit distance, all relative to rail transport. Standard errors corrected for clustering at the destination district level are reported in parentheses of column 1, and bootstrapped 95 percent confidence intervals are reported in column 2.

TABLE 3—RAILROADS AND TRADE FLOWS: STEP 2

Dependent variable: log value of exports	(1)	(2)
log effective distance between origin and destination along lowest-cost route	-1.603 (0.533)	-1.701 (1.141)
(log effective distance between origin and destination along lowest-cost route) $\times$ (weight per unit value of commodity in 1890)		-0.946 (3.634)
(log effective distance between origin and destination along lowest-cost route) $\times$ (high-value railroad freight class of commodity in 1859)		1.286 (1.243)
Observations	142,541	142,541
R^2	0.901	0.901

Notes: Regressions estimating equation (14) using data on 15 commodities and 47 trade blocks annually from 1882 to 1920. Regressions include origin and destination fixed effects, separately for each commodity and year. "Effective distance between origin and destination along lowest-cost route" measures the railroad-equivalent kilometers (due to the normalization of railroad distance cost to 1) between the centroid of the origin and destination trade blocks in question, along the lowest-cost route given relative freight rates for each mode of transport (as estimated in Table 2). "Weight per unit value in 1890" is the weight (in maunds) per rupee, as measured by 1890 prices. "Railroad freight class in 1859" is an indicator variable for all commodities that were classified in the higher (more expensive) freight class in 1859; salt is in the omitted category (low-value commodities). Heteroskedasticity robust standard errors adjusted for clustering at the exporter-importer block level are reported in parentheses for columns 1 and 2 respectively.

(a) Table 2: Railroads and trade costs.

(b) Table 3: Railroads and trade flows.

Tables 2 and 3: Steps 1 and 2 of the empirical mechanism.

6.4 Tables 4 and 5: Real income and sufficient statistic

Read:

- Table 4 shows that railroad access increases log real agricultural income by 0.164 in the baseline specification, implying roughly a 16 percent increase.
- The placebo-line specifications do not show similar effects for unbuilt lines, including lines abandoned after proposal, reconnaissance, or survey stages and lines from alternative historical plans.
- Table 5 adds the model-based trade share sufficient statistic. The coefficient on railroad access falls from 0.258 to 0.124.
- The model-computed trade share has a strong negative coefficient of -1.587 , consistent with the idea that districts become richer when they rely less on self-sourcing and gain more from trade.
- The decline in the railroad coefficient implies that the trade-cost channel accounts for a little over one-half of the reduced-form railroad income effect.

Economic message: These tables connect infrastructure to welfare. Railroads raise real income, and a large part of this effect operates through reduced trade costs and expanded gains from trade.

TABLE 4—RAILROADS AND REAL INCOME LEVELS: STEP 3

Dependent variable: log real agricultural income	(1)	(2)	(3)	(4)
Railroad in district	0.164 (0.049)	0.158 (0.048)	0.160 (0.050)	0.167 (0.050)
Unbuilt railroad in district, abandoned after proposal stage		0.057 (0.058)		
Unbuilt railroad in district, abandoned after reconnaissance stage		0.013 (0.099)		
Unbuilt railroad in district, abandoned after survey stage		−0.069 (0.038)		
(Unbuilt railroad in district, included in Lawrence Plan 1869–1873) × (post-1871 indicator)			0.067 (0.104)	
(Unbuilt railroad in district, included in Lawrence Plan 1874–1878) × (post-1874 indicator)			−0.019 (0.092)	
(Unbuilt railroad in district, included in Lawrence Plan 1879–1883) × (post-1879 indicator)			0.095 (0.084)	
(Unbuilt railroad in district, included in Lawrence Plan 1884–1888) × (post-1884 indicator)			−0.072 (0.075)	
(Unbuilt railroad in district, included in Lawrence Plan 1889–1893) × (post-1889 indicator)			0.047 (0.049)	
(Unbuilt railroad in district, included in Lawrence Plan 1894–1898) × (post-1894 indicator)			−0.088 (0.086)	
(Unbuilt railroad in district, included in Kennedy plan, high-priority) × (year-1848)			−0.0001 (0.002)	
(Unbuilt railroad in district, included in Kennedy plan, low-priority) × (year-1848)			0.001 (0.003)	
Observations	7,086	7,086	7,086	7,086
R ²	0.848	0.848	0.848	0.848

Notes: OLS regressions estimating equation (16) using real income constructed from crop-level data on 17 principal agricultural crops (listed in online Appendix A), from 192 districts in India, annually from 1870 to 1930. All regressions include district fixed effects and year fixed effects. “Railroad in district” is a dummy variable whose value is 1 if any part of the district in question is penetrated by a railroad line. “Unbuilt railroad in district, abandoned after X stage” is a dummy variable whose value is 1 if a line that was abandoned after “X” stage penetrates a district, in all years after the line was first mentioned as reaching stage “X” in official documents. Stages “X” are: “proposal,” where the line was mentioned in official documents; “reconnaissance,” where the line route was explored by surveyors in rough detail; and “survey,” where the exact route of the line and nature of all engineering works were decided on after detailed survey. “Lawrence 1868 plan” was a proposal for significant railroad expansion by India’s Governor General that was not implemented; the plan detailed proposed dates of construction (in 5-year segments) over the next 30 years, which are used in the construction of this variable. “Kennedy plan” was an early construction-cost minimizing routes plan drawn up by India’s chief engineer in 1848 (divided into high- and low-priorities), which was rejected in favor of Dalhousie’s direct routes plan. Heteroskedasticity-robust standard errors corrected for clustering at the district level are reported in parentheses.

TABLE 5—A SUFFICIENT STATISTIC FOR RAILROAD IMPACT: STEP 4

log real ag. income, corrected for rainfall:	(1)	(2)
Railroad in district	0.258 (0.050)	0.124 (0.050)
“Trade share,” as computed in model		−1.587 (0.177)
Observations	7,086	7,086
R ²	0.835	0.844

Notes: OLS Regressions estimating equation (18) using real income constructed from crop-level data on 17 principal agricultural crops (listed in online Appendix A), from 192 districts in India, annually from 1870 to 1930. Dependent variable is log real income, corrected for crop-specific rainfall of each of 17 crops, weighted across crops as in equation (18). Regressions include district fixed effects and year fixed effects. “Railroad in district” is a dummy variable whose value is one if any part of the district in question is penetrated by a railroad line. “Trade share” is the share of a district’s expenditure that it buys from itself; this variable is computed in the equilibrium of the model, where the model parameters are set to those estimated in Steps 1 and 2, and the exogenous variables (the transportation network, rainfall, and district land sizes) are as observed. Heteroskedasticity-robust standard errors corrected for clustering at the district level are reported in parentheses.

(a) Table 4: Real income and placebo checks.

(b) Table 5: Sufficient statistic.

Tables 4 and 5: Steps 3 and 4 of the empirical mechanism.

7 Short summary

- The paper estimates the economic impact of colonial India’s railroad network.
- It constructs a new archival dataset on district output, prices, rainfall, internal and external trade flows, and railroad construction dates.
- A Ricardian trade model gives four empirical steps: estimate trade costs, estimate trade flows, estimate real-income effects, and test a sufficient statistic for welfare.
- Salt price gaps show that railroads reduced trade costs by lowering effective distance across a multimodal transport network.
- Gravity estimates show that lower effective distance increased trade flows.
- Reduced-form estimates show that railroad access raised real agricultural income per acre by about 16 percent.
- Placebo lines that were planned or surveyed but never built do not generate comparable income effects.

- The model-based own-trade share explains slightly more than one-half of the reduced-form railroad income effect.
- The paper’s main contribution is to connect infrastructure, trade costs, trade flows, and welfare in one coherent empirical and theoretical framework.

8 Conclusion

8.1 Core conclusion

Donaldson shows that railroads had large economic effects in colonial India. They lowered trade costs, narrowed spatial price gaps, expanded interregional and international trade, and raised real agricultural income. The headline reduced-form estimate is that district railroad access increased real agricultural income by roughly 16 percent.

8.2 Economic intuition

The economic intuition is that railroads transformed local economies from relatively isolated production-consumption units into regions connected to a broader market. Once transport costs fall, a district can sell more of the goods it is relatively good at producing and buy more of the goods that other districts can produce cheaply. The gains arise not simply because moving goods becomes cheaper, but because lower trade costs allow comparative advantage to become economically relevant.

8.3 Incremental contribution revisited

The paper advances the infrastructure and trade literatures in three connected ways. First, it constructs unusually rich historical data that allow the paper to observe prices, trade, rainfall, output, and network expansion together. Second, it introduces a network-based method for estimating trade costs from unique-source commodity price gaps. Third, it uses a general equilibrium trade model not merely for calibration, but to discipline the empirical mechanism and assess how much of the reduced-form income gain is due to trade.

8.4 Implications for infrastructure and trade policy

The findings imply that transportation infrastructure should be evaluated not only by engineering cost savings but also by its effect on market integration and welfare. Infrastructure that lowers trade costs can raise real income by allowing regions to specialize, trade, and diversify their sources of goods. At the same time, the fact that the trade-share mechanism explains only about one-half of the reduced-form effect suggests that infrastructure may also matter through other channels such as information diffusion, technology adoption, factor mobility, public administration, or risk sharing.

8.5 Limitations

The outcome variable is real agricultural income per acre, not a full measure of household welfare or aggregate GDP. The railroad treatment may include bundled improvements in communication and administrative integration, especially telegraph and postal services. The structural mechanism depends on the assumptions of the Eaton-Kortum-style model, the use of rainfall to measure productivity, and the computed trade share. Finally, the paper focuses on income levels and says less about volatility, famine risk, and distributional consequences.

8.6 Takeaway

The enduring takeaway is that infrastructure matters because it changes the economic geography of trade. Donaldson's central achievement is to show, using historical data and a model-guided empirical design, that railroads raised income in colonial India substantially, and that a large share of this gain can be traced to lower trade costs and the resulting gains from trade.